

ALTIRIG · DESIGN DECISIONS REGISTER

AltiRig design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/altirig		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Amish's decisions 41B (80 % open baffle) and 42B (surplus pipe and heads, host lab's DC supply and pump; R10 restated) moved to Decisions made (ALR-DDR-003)
0.1	2026-10-03	Amish Chadha	Register opened at TRL 3; design decisions made under Amish's 2026-10-03 pre-approvals; R6 and R10 posed to Amish

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands.

Safety: AltiRig is a vacuum vessel with a spinning propeller inside, run from mains power. Every decision that touches safety took the conservative option, and the evidence that would relax it is named in its record. Nothing here makes the chamber certified pressure or test equipment.

■ Open decisions

None. The two requirement decisions put to Amish at TRL 3 (R6 and R10) were decided on 2026-10-03 as recommended (41B and 42B, below and in [ALR-DDR-003](#)); carrying them out raised no new question for Amish. What the host lab's pump and supply must meet is listed under To confirm.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The surplus pipe offcut's out-of-roundness (no more than 8 mm between largest and smallest diameter) and wall thickness (6.35 mm, no less than 5.9 mm at any point), checked before buying	The shell's collapse factor of 6.7 assumes both; decision 42B makes the pipe surplus	ALR-CAL-001 B; ALR-DDR-003
2	The surplus or overstock tank heads' skirt length (40 mm), thickness (no less than 5.9 mm) and outside diameter match the pipe, checked before buying	The door ring fits over the skirt; the rear head butts to the pipe	ALR-DDR-002, C2; ALR-DDR-003
3	Flatness of both flange ring faces after welding (within 1 mm)	The 6 mm sponge must seal the door at the first pump-down	ALR-BLD-001, 3.2
4	The relief valve's adjustable range covers 55 kPa below atmosphere and it is rated for vacuum service	R12	ALR-DDR-001, D4
5	The load cells' stiffness (deflection at full scale) and combined error class	Sets the share of thrust carried by the leaves (2.9 % assumed) and the error budgets	ALR-CAL-001 E
6	The host lab's pump: rated at least 140 L/min (170 L/min as calculated), with a thermal overload and an oil-mist exhaust filter; its speed curve between 50 and 100 kPa inlet	Pump-down time (R8) assumes 85 % of the rating; 140 L/min gives about 4.9 min against 5 min	ALR-CAL-001 C and H; ALR-DDR-003
7	The example propeller's static coefficients (0.10 and 0.042 assumed)	Speeds, power and torque at 50 N; the supply limit of about 44 N at 5,000 m density	ALR-CAL-001 E
8	Latch clamps hold at least 3 kN	The first seal needs about 2.1 kN each	ALR-CAL-001 B
9	The host lab's DC supply gives at least 1,500 W at 24 V (adjustable 20 to 28 V), has overcurrent protection, and can be switched by the cabinet's motor contactor	A smaller supply lowers the thrust reached at 5,000 m density below about 44 N; R11 needs the contactor to cut it	ALR-CAL-001 E and H; ALR-DDR-003

■ Value engineering

Value-engineering target: USD 1,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 3,865 (USD 2,865 over the target). R10 is restated to USD 4,000 by Amish's decision 42B, and USD 3,865 is USD 135 under it. Main cost drivers and savings still worth trying:

- The vessel steel, heads and nozzles (about USD 1,270 with the surplus pipe and heads, saddles, rings, rails, tabs and window rings) and the welding labour (USD 650). A used air receiver of the same size with its own heads is the next saving worth pricing.
- The speed controller and power sensor (USD 100) and the safety controls and cabinet (USD 360). The host lab's DC supply is used (decision 42B), keeping the interlocked contactor.
- The valve manifold (USD 220) and hose (USD 45). The host lab's pump is used (decision 42B).
- The thrust stand itself is cheap (about USD 300 including the load cells); it is not where the money goes.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	TRL 2 review items D1 to D14: match density not pressure, pipe and tank-head vessel, design for full vacuum, R12 relief valve, R11 interlocks, polycarbonate window outside the fragment zone, flexure stand, room temperature only, removable baffle with comparison runs, no batteries inside, external motor supply, propeller kit left to Lift and Range, shared blocks, not-certified notice with engineer review and proof pump-down	Amish: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." and "Proceed with the remaining 15 scaffolds"	ALR-DDR-001
2026-10-03	Co-design: the first candidate partner to approach is the Department of Aerospace Engineering, IIT Kanpur, then Kathmandu University; first candidate region the Nepal and Indian Himalaya; candidates, not agreed	Amish, same pre-approvals	ALR-DDR-001 , D11
2026-10-03	Design for construction C1 to C17: pipe shell, flat flange rings with a sponge gasket, set-in heavy-wall nozzles, window stack with a spacer ring, potted feed-through plate, saddles, rails and bolted deck, bolted baffle on tabs, hinge and latches, flexure stand with an upright cell, torque head on bearings, speed post, manifold on a stub flange, normally-open bleed valve, feed-through placement, pump and cabinet placement	Amish, same pre-approvals	ALR-DDR-002
2026-10-03	Cost overrun against the value-engineering target accepted; budget_usd left at USD 1,000	Amish, same pre-approval ("I also accept any cost overruns")	This register, Value engineering
2026-10-03	41B, R6: an 80 % open baffle (2 mm sheet, 22 mm square holes on a 24.5 mm pitch) on the same tabs; chamber effect now 1.8 % to 3.3 % against 5 %; the 1,016 mm vessel held in reserve if the TRL 4 comparison shows more than 5 %	Amish, 2026-10-03: "i agree with all the 46 recommendations you provided. please proceed."	ALR-DDR-003
2026-10-03	42B, R10: surplus pipe offcut and tank heads (after the roundness and thickness checks) and the host lab's DC supply and vacuum pump; cost USD 3,865; R10 restated to "complete rig at or below USD 4,000 with surplus pipe and heads and the host lab's DC supply and pump"	Amish, 2026-10-03: "i agree with all the 46 recommendations you provided. please proceed."	ALR-DDR-003